STA-100

General math & Bio stat

| Handouts |

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STA301 Statistics and Probability

Final Term Examination - August 2004

Time Allowed: 150 Minutes

Instructions

Please read the following instructions carefully before attempting any of the questions:

1. Attempt all questions. Marks are written adjacent to each question.
2. Do not ask any questions about the contents of this examination from anyone.
   a. If you think that there is something wrong with any of the questions, attempt it to the best of your understanding.
   b. If you believe that some essential piece of information is missing, make an appropriate assumption and use it to solve the problem.
   c. Write all steps, missing steps may lead to deduction of marks.
3. You are allowed to use the calculator & Statistical tables in order to solve the questions.
4. For your convenience we are providing you the following symbols,

□ ,?, $\bar{X}$ or write Mean, s, ∫ or sd for standard deviation, s2 ∫2 or sd2 or variance for variance, $\sqrt{\ }$ for square root or whole square root.

****WARNING: Please note that Virtual University takes serious note of unfair means. Anyone found involved in cheating will get an `F` grade in this course.**

Total Marks: 50 **Total Questions: 15**

Question No. 1 **Marks : 3**

Write down any three types of average.

Question No. 2 **Marks : 3**

Define the Trial and the Outcome.

Question No. 3 **Marks : 3**

What is meant by Sampling Distribution?

Question No. 4 **Marks : 3**

Define the following:

1. Null Hypothesis
2. Alternative Hypothesis

Question No. 5 **Marks : 1**

Which of the following is the example of discrete variable?

- The number of children in different families of a locality
- Ages of cancer patients
- Height of students
- None of these

Question No. 6 **Marks : 1**

The Mode of the letters in the word STATISTICS is:

- S
- T
- I
- S and T

Question No. 7 **Marks : 1**

In Binomial Experiment, successive trails are:

- Both independent and dependent.
- Dependent
- Independent
- None of above

Question No: 1 (Marks: 1) - Please choose one

For a positively skewed distribution m_3 will be:

- ► Positive
- ► Negative
- ► Zero
- ► 1

Question No: 2 (Marks: 1) - Please choose one

When data is labeled to identify an attribute of element, the measurement scale is:

- ► Ordinal
- ► Interval
- ► Nominal
- ► Ratio

Question No: 3 (Marks: 1) - Please choose one

Suppose the estimated equation is $\hat{Y} = 5 - 2X$ has been calculated for a set of data. What is slop of the line:

- ► 0
- ► 2
- ► -2
- ► 5

Question No: 4 (Marks: 1) - Please choose one

If P(B|A) = 0.25 and $P(A \cap B) = 0.20$, then P(A) is:

- ► 0.05
- ► 0.80
- ► 0.95
- ► 0.75

Question No: 5 (Marks: 1) - Please choose one

Which branch of statistics deals with the techniques that are used to organize, summarize, and present the data:

- ► Advance statistics
- ► Probability statistics
- ► Descriptive statistics
- ► Inferential statistics

Question No: 6 (Marks: 1) - Please choose one

In a sample of 800 students in a university, 160, or 20%, are Business majors. Based on the above information, the school's paper reported that "20% of all the students at the university are Business majors." This report is an example of :

- ► A sample
- ► A population
- ► Sstatistical inference
- ► Descriptive statistics

Question No: 7 (Marks: 1) - Please choose one

A set that contains all possible outcomes of a system is known as

- ► Finite Set
- ► Infinite Set
- ► Universal Set
- ► No of these

Question No: 8 (Marks: 1) - Please choose one

If X and Y are independent, then Var(X-Y) is equal to:

- ► $Var(X)-Var(Y)$
- ► $Var(X)+Var(Y)$
- ► $Var(X+Y)$
- ► Zero

Question No: 9 (Marks: 1) - Please choose one

Which of the following is the class frequency

- ► The number of observations in each class
- ► The difference between consecutive lower class limits
- ► Always contains at least 5 observations
- ► Usually a multiple of the lower limit of the first class

Question No: 10 (Marks: 1) - Please choose one

How to construct the class interval:

- ► Divide the class frequencies in half
- ► Divide the class frequency by the number of observations
- ► Find the difference between consecutive lower class limits
- ► Count the number of observations in the class

Question No: 11 (Marks: 1) - Please choose one

Data in the Population Census Report is:

► Ungrouped data

► Secondary data

► Primary data

► Arrayed data

Question No: 12 (Marks: 1) - Please choose one

What is the range of -2,-3,-5,-10 :

► -12

► 8

► -8

► 2

Question No: 13 (Marks: 1) - Please choose one

The algebraic sum of deviations from mean is:

► Maximum

► Minimum

► Zero

► Undefined

Question No: 14 (Marks: 1) - Please choose one

The sum of squares of deviations from mean is:

► Undefined

► Zero

► Maximum

► Minimum

Question No: 15 (Marks: 1) - Please choose one

Statistic is a numerical quantity, which is calculated from:

► Population

► Sample

► Data

► Observations

Question No: 16 (Marks: 1) - Please choose one

Which of the following is not based on all the observations?

- Arithmetic Mean
- Geometric Mean
- Harmonic mean
- Mode

Question No: 17 (Marks: 1)

Elaborate the word dispersion.

We can say that the degree of scatter of data, usually about an average value, can be the median.

Question No: 18 (Marks: 1)

Define population.

We can define population is the collection of individuals or objects having some common measurable characteristics.

Question No: 19 (Marks: 2)

What does mean by the independence of two events:

We can define independence of two events are statistically independent if the probability of their occurring jointly equals the product of their respective probabilities. Independence of two events also know as stochastic independence.

Question No: 20 (Marks: 3)

The reciprocal of the values are

0.012, 0.0235, 0.0135

Calculate Harmonic Mean

Harmonic mean is 250.037

Question No: 21 (Marks: 5)

The probability that a student passes mathematics is 2/3 and the probability that he passes English is 4/9. If the probability of passing at least one course is 4/5, what is the probability that he will pass both courses?

Math=2/3
English=4/9
Least one passing probability= 4/5

Math + English =2/3+4/9
=1.11
4/5+1.11

=1.911

Question No: 22 (Marks: 10)

A pair of dice is thrown, then

1) Find the sample space for this experiment

Suppose if A and B is the pair of dice then lets try to find out how many result we can find

- A hit first
- B Hit first
- B hit Second
- A Hit Second
- Both Hit Equally
- Both didn't Hit

2) Determine the probability of getting the sum 8 on the dice

3) Find the probability of getting sum 7 or 11

For Teacher's use only											
Question Marks	1	2	3	4	5	6	7	8	9	10	Total
Question Marks	11	12	13	14							

Question No: 1 (Marks: 4)

C.V of scores made by two batsmen A and B in a series of innings are C.VA= 117.67% and C.VB=70.45%. Who is more consistent player?

Question No: 2 (Marks: 4)

What is meant by estimation? What are its types?

Question No: 3 (Marks: 4)

Write down any four properties of normal distribution.

Question No: 4 (Marks: 1) - Please choose one

For 9 observations all consisting of 4, the following relation between A.M, G.M and H.M. will hold:

- A.M.>G.M.>H.M
- A.M. <G.M. <H.M
- A.M. =G.M. =H.M.
- None of these

Question No: 5 (Marks: 1) - Please choose one

If $P\left(A*B\right)=P\left(A\right)+P\left(B\right),$ **then A and B are:**

► Mutually exclusive

► Dependent

► Independent

► None of these

Question No: 6 (Marks: 1) - Please choose one

In which distribution the successive trials are without replacement:

► Hypergeometric distribution

► Binomial distribution

► Continuous distribution

► None of these

Question No: 7 (Marks: 1) - Please choose one

A _____ is a subset of a _____.

► Sample, population

► Population, sample

► Statistic, parameter

► Parameter, statistic

Question No: 8 (Marks: 1) - Please choose one

If false hypothesis is accepted, it is called :

- ► Level of significance
- ► Type-I error
- ► Type-II error
- ► None of these

Question No: 9 (Marks: 1) - Please choose one

The points of inflection in normal distribution are:

- ► $\mu-\sigma, \mu+\sigma$
- ► $\mu-2\sigma, \mu+2\sigma xz$
- ► μ, σ
- ► None of these

Question No: 10 (Marks: 1) - Please choose one

For testing of hypothesis about population proportion, we use:

- ► Z-test
- ► t-test

► Chi-square test

► None of these

Question No: 11 (Marks: 1) - Please choose one

Which of the following cannot be considered as null hypothesis (H_0) ?

► $\backslash = \backslash_0$

► $\backslash \delta \backslash_0$

► $\backslash \varepsilon \backslash_0$

► $\backslash > \backslash_0$

Question No: 12 (Marks: 10)

Calculate the Geometric Mean.

Marks	Number Of Students
30-39	8
40-49	87
50-59	190
60-69	304
70-79	211
80-89	85
90-99	20

Question No: 13 (Marks: 10)

A research worker wishes to estimate the mean of a Population using a sample sufficiently large

that the probability will be 0.95 that the sample mean will not differ from the true mean by more than 25 pr cent of the standard deviation. How large a sample should be taken?

Question No: 14 (Marks: 10)

Random samples of 200 bolts manufactured by machine A showed 19 and 100 bolts manufactured by machine B showed 5 defective bolts. Test the hypothesis at 5% level of significance that the two machines are showing different qualities of performance.

Question No. 8 **Marks : 1**

In a Binomial Distribution n = 10, p = 3/5. its mean will be:

- o 12/5
- o 1
- o 2/5
- o 6

Question No. 9 **Marks : 1**

A population that consists of unlimited number of elements is called:

- o Finite population
- o Infinite population
- o Hypothetical population
- o Target population

Question No. 10 **Marks : 1**

An estimator which is free from Bias is called:

- o Biased Estimator
- o Unbiased Estimator
- o Consistent Estimator
- o Efficient Estimator

Question No. 11 **Marks : 1**

The probability of accepting the true null hypothesis is called:

- o Level of significance
- o Level of confidence
- o Type-I Error
- o Type-II Error

Question No. 12 **Marks : 1**

If the population standard deviation is not known and sample size is large (n > 30), then the test statistic to be used is:

- o t-test
- o Z-test
- o F-test
- o χ^2-test

Question No. 13 **Marks : 10**

Calculate the Mean Deviation from mean for the following data.

Size of items	3-4	4-5	5-6	6-7	7-8
Frequency	3	7	22	60	8

Question No. 14 **Marks : 10**

The probability that a patient recovers from a delicate heart operation is 0.9. What is the probability that exactly five of the next 7 patients having this operation survive.(Use Binomial Distribution)

Question No. 15 **Marks : 10**

A standard examination has been given for several years with μ = 70 and σ^2 = 9.A school using this examination for the first time, gave it to a group of 25 students who obtained a mean $\bar{X} = 71$ and a variance of $S^2 = 12$. Test the hypothesis that $H_0 : \sigma^2 = 9$ against the alternative $H_1 : \sigma^2 \neq 9$ at 5% level of significance.

For Teacher's use only

Questions	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8
Marks								

Questions	Q9	Q10	Q11	Total
Marks				

Question No: 1 **Marks:1**

If mean=median=mode then curve is symmetric.

- o True
- o False

Question No: 2 **Marks:1**

Distribution Function is also called the step function.

- o True
- o False

Question No: 3 **Marks:1**

In the case of sampling with replacement as well as in the case of sampling without replacement, we have:

$\mu_{\bar{X}} = \mu$

- o True
- o False

Question No: 4 **Marks:1**

The rejection region is also called the critical region.

- o True
- o False

Question No: 5 **Marks:4**

Describe the continuous and discrete random variable with example.

Question No: 6 **Marks:4**

Define and describe the Least Significance Difference (LSD).

Question No: 7 **Marks:4**

Write down the properties of normal distribution.

Question No: 8 **Marks:4**

Discuss on null and alternative hypothesis.

Question No: 9 **Marks:10**

Following is given the data of Traffic Density and Accident rate.

Traffic Density	30	35	40	45	50	60	70	80	90
Accident rate	2	4	5	5	8	15	24	30	32

a. Plot the data on scatter dia-gram

b.Calculate the correlation co-efficient between traffic density and accident rate

c.Interpret the correlation.

Question No: 10 **Marks:10**

A continuous random variable X that can assume values between x=1 and x=4 has a density function given by

$f(x) = \frac{1}{3}$

a. Show that the area under the curve is equal to 1.

b.Find $p(1.5 < X < 3)$

c. $P(X \leq 2.2)$

Question No: 11 **Marks:10**

Let X and Y be two discrete r.vs with the following joint p.d.

x **y**	2	4
1	0.1	0.15
3	0.2	0.30
5	0.1	0.15

Find $E(X)$ and $E(Y)$.

	FINALTERM EXAMINATION FALL 2006 STA301 - STATISTICS AND PROBABILITY (Session - 1)	Marks: 50 Time: 120min

StudentID/LoginID: _____

Student Name: _____

Center Name/Code: _____

Exam Date: Tuesday, February 06, 2007

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3. You are allowed to use the calculator & Statistical tables in order to solve the questions.

4. For your convenience we are providing you the following symbols,

Σ **,**$\cap$**,** $\bar{X}$ **o**r write Mean, **s,**σ or **sd** for standard deviation, **s²** σ^2 or **sd²** or variance for variance, $\sqrt{\ }$, $\sum \log x$,for square root or whole square root.

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For Teacher's use only											
Question	1	2	3	4	5	6	7	8	9	10	Total
Marks											
Question	11	12	13	14							
Marks											

Question No: 1 (Marks: 4)

Statistics as a subject, in which two of parts is divided? Expalin briefly both of parts.

Question No: 2 (Marks: 4)

Differentiate simple and composite hypothesis.

Question No: 3 (Marks: 4)

Correct the followings:

$\mu \pm \sigma$ contains approximately 50% area.

$\mu \pm 2\sigma$ contains approximately 90% area.

$\mu \pm 3\sigma$ contains approximately 90.88% area.

Question No: 4 (Marks: 1) - Please choose one

The heights in centimeters of 5 students are:
165, 175, 176, 159, 170.
The sample median and sample mean are respectively:

- 170, 169
- 170, 170
- 169, 170
- 176, 169

Question No: 5 (Marks: 1) - Please choose one

The characteristic which can not be measured numerically is called:

- Quantitative variable
- Qualitative variable
- Discrete variable

- Continuous variable

Question No: 6 (Marks: 1) - Please choose one

The expected value of the normal distribution is

- 0
- 1
- μ
- σ

Question No: 7 (Marks: 1) - Please choose one

Normal distribution is

- Uni-model
- Bi-modal
- Multi-model
- None of these

Question No: 8 (Marks: 1) - Please choose one

One sided and two sided critical regions are based on:

- Level of significance
- Sample size

- Null hypothesis
- Alternative hypothesis

Question No: 9 (Marks: 1) - Please choose one

The rule or formula that is used to estimate a population parameter is called:

- Estimate
- Estimator
- Denominator
- None of these

Question No: 10 (Marks: 1) - Please choose one

The probability of rejecting a true null hypothesis is called:

- Level of significance
- Type-1 error
- Type-II error
- None of above

Question No: 11 (Marks: 1) - Please choose one

The value of chi-square can never be

- Zero
- Negative
- Greater than 1
- None of these

Question No: 12 (Marks: 10)

The grade-point averages of college seniors selected at random from the graduating class are as follc

3.2	1.9	2.7	2.4
2.8	2.9	3.8	3.0
2.5	3.3	1.8	2.5
3.7	2.8	2.0	3.2
2.3	2.1	2.5	1.0

Calculate the standard deviation.

Question No: 13 (Marks: 10)

The mean lifetime of electric light bulbs produced by a company has in the past been 1120 hours with a standard deviation of 125 hours. A sample of 8 electric bulbs recently chosen form a supply of newly manufactured bulbs showed a mean lifetime of 1070 hours. Test the hypothesis that mean lifetime of the bulbs has not changed using a level of significance of 0.05.

Question No: 14 (Marks: 10)

A random sample of 200 voters is selected and 120 are found to support an annexation suit. Find the 96% confidence interval for the fraction of the voting population favoring the suit.

Marks											
Question	11	12	13	14							
Marks											

Question No: 1 (Marks: 4)

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Differentiate simple and composite hypothesis.

Question No: 3 (Marks: 4)

Correct the followings:

$\propto \pm \int$ contains approximately 50% area.

$\propto \pm 2 \int$ contains approximately 90% area.

$\propto \pm 3 \int$ contains approximately 90.88% area.

Question No: 4 (Marks: 1) - Please choose one

The heights in centimeters of 5 students are:
165, 175, 176, 159, 170.
The sample median and sample mean are respectively:

- ▶ 170, 169
- ▶ 170, 170
- ▶ 169, 170
- ▶ 176, 169

Question No: 5 (Marks: 1) - Please choose one

The characteristic which can not be measured numerically is called:

- ▶ Quantitative variable

► Qualitative variable

► Discrete variable

► Continuous variable

Question No: 6 (Marks: 1) - Please choose one

The expected value of the normal distribution is

► 0

► 1

► $\propto$

► $\int$

Question No: 7 (Marks: 1) - Please choose one

Normal distribution is

► Uni-model

► Bi-modal

► Multi-model

► None of these

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- Estimator
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- None of these

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- Level of significance
- Type-1 error

► Type-II error

► None of above

Question No: 11 (Marks: 1) - Please choose one

The value of chi-square can never be

► Zero

► Negative

► Greater than 1

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The grade-point averages of college seniors selected at random from the graduating class are as foll o

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2.8	2.9	3.8	3.0
2.5	3.3	1.8	2.5
3.7	2.8	2.0	3.2
2.3	2.1	2.5	1.0

Calculate the standard deviation.

Question No: 13 (Marks: 10)

The mean lifetime of electric light bulbs produced by a company has in the past been 1120 hours with a standard deviation of 125 hours. A sample of 8 electric bulbs recently chosen form a supply of newly manufactured bulbs showed a mean lifetime of 1070 hours. Test the hypothesis that mean lifetime of the bulbs has not changed using a level of significance of 0.05.

Question No: 14 (Marks: 10)

A random sample of 200 voters is selected and 120 are found to support an annexation suit. Find the 96% confidence interval for the fraction of the voting population favoring the suit.

If $m_1 = 0$, $m_2 = 2.64$, $m_3 = 0.08$, $m_4 = 28.30$ find b_2.

Question No: 2 (Marks: 4)

What is (1-α) .Explain it?

Question No: 3 (Marks: 4)

Differentiate between simple and composite hypothesis.

Question No: 4 (Marks: 1) - Please choose one

Correction factor $\frac{N-n}{N-1}$ is used for:

- n is small
- n is large
- Sampling without replacement
- Sampling with replacement

Question No: 5 (Marks: 1) - Please choose one

In Binomial Distribution, the random variable "X" has a range:

- 0, 1, 2, …, n
- 0, 1, 2, …, ∞
- - ∞ to + ∞
- - ∞ to 0

Question No: 6 (Marks: 1) - Please choose one

Probability of a 'Jack card' from the 52 playing cards is:

- $\frac{1}{52}$
- $\frac{4}{52}$
- $\frac{13}{52}$
- None of these

Question No: 7 (Marks: 1) - Please choose one

In normal distribution

- Mean > Median > Mode
- Mean<Median< Mode
- Mean= Median=Mode
- None of the these

Question No: 8 (Marks: 1) - Please choose one

If a false hypothesis is accepted, it is called:

- Level of significance
- Type-I error
- Type-II error
- Level of confidence

Question No: 9 (Marks: 1) - Please choose one

Which of the following is not composite hypothesis?

- $H_0 : \theta = \theta_0$
- $H_0 : \theta \leq \theta_0$
- $H_0 : \theta \geq \theta_0$
- $H_0 : \theta \neq \theta_0$

Question No: 10 (Marks: 1) - Please choose one

If the population standard deviation is not known and the sample size is large $(n \geq 30)$, then the test statistic to be used is

- t-test
- Z-test
- Chi-square test
- None of these

Question No: 11 (Marks: 1) - Please choose one

Critical region is the part of the sampling distribution for which the null hypothesis is

- Rejected
- Accepted
- Ignored
- None of these

Question No: 12 (Marks: 10)

The table shows the frequency distribution of the number of spelling mistakes in a composition made by each pupil in class of 36.

No of Mistakes (x)	No of Pupils (f)

0	3
1	7
2	10
3	6
4	5
5	3
6	1
7	1

Calculate the Mean, Median and Mode.

Question No: 13 (Marks: 10)

A secretary makes 2 errors per page on the average. What is the probability that on the next page she makes

(a) 4 or more errors
(b) No error

Question No: 14 (Marks: 10)

A producer of a certain flashlight dry cell batteries claims that its output has a mean life of 750 minutes. A random sample of 15 such batteries was tested and a sample mean of 745 minutes with a sample s.d of 24 minutes was obtained. Verify that these results are consistent with the null hypothesis $\mu \geq 750$ against $\mu < 750$ at Use. $\alpha = 0.01$